LESSON PLAN

NBCD (30)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	04
Total No. of Lesson Plans	6	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Operation of oscilloscope and function generator and its controls and observe waveform, measurement of various parameters of waveform.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Operation of oscilloscope and function generator and its controls and observe waveform, measurement of various parameters of waveform.
5. Specific Objectives	30.1.1.1 Trainee should be able to practical on various lab FTL, RF & Embedded lab.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Operation of oscilloscope and function generator, observation of waveforms, measurement of amplitude, frequency, and period.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of an oscilloscope?
(b) Define the term 'waveform' in the context of electronics.
(c) List two common types of waveforms used in signal generation.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Introduction to Oscilloscope and Function Generator • Identification of Front Panel Controls • Power On/Off and Initial Setup Procedures • Selection of Probe Attenuation and Grounding • Basic Waveform Display and Triggering	L & Ds	PPT, IB	15 min
(b)	• Generation of Sine, Square, and Triangle Waves • Adjustment of Amplitude and Frequency Controls • Measurement of Peak-to-Peak Voltage • Measurement of Period and Frequency Calculation • Use of Horizontal and Vertical Scale Factors	L & Ds	PPT, PDF	20 min
(c)	• Troubleshooting Common Display Issues • Verification of Measurement Accuracy • Safety Precautions During Practical Session • Documentation of Observed Waveforms • Instructor Supervision and Feedback	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between the horizontal time base and the measured frequency.
(b) Describe the steps required to stabilize a waveform on the oscilloscope screen.
(c) How do you calculate the peak-to-peak voltage from the vertical scale setting?

12. Home Assignment and Follow Up	Prepare a brief report on the operation of an oscilloscope and function generator to be submitted before LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Construct, test and observe output on voice transmission using amplitude modulation and demodulation on optical fiber communication trainer.	Refer to the master syllabus reference list	LP 05

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min